# Mallavarapu Naveen Kumar

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GitHub | LinkedIn | Portfolio

## Professional Summary

Computer Science and Engineering graduate with a strong foundation in Python, web development, database management, and machine learning. Experienced in developing full-stack applications and AI-powered solutions through hands-on academic projects and a technical internship. Familiar with REST API development, RAG, resume analysis, and machine learning integration. A proactive learner with strong problem-solving abilities and a passion for building scalable, technology-driven solutions. Seeking an entry-level software development opportunity to apply technical knowledge and contribute to innovative projects.

## Education

**Marri Laxman Reddy Institute of Technology and Management** **2022 – 2026**
*B.Tech – Computer Science & Engineering* *CGPA: 7.20*

## Experience

**Deep Learning Intern** **01 Aug 2024 – 14 Aug 2024**
*Central Institute of Tool Design (CITD)*

- Gained hands-on experience in deep learning methodologies, neural network architectures, and AI model development.
- Performed data preprocessing and explored model training, validation, and performance evaluation using Python-based deep learning frameworks.
- Developed practical understanding of deep learning workflows and their applications in solving real-world problems.

## Projects

**Personal Health Assistant Chatbot** *GitHub | Individual*
*React.js, FastAPI, RAG, Groq API, LLM, ChromaDB, MongoDB Atlas, Python*

- Developed an AI-powered healthcare chatbot using Retrieval-Augmented Generation (RAG) to provide context-aware responses to general health-related queries.
- Implemented semantic search using ChromaDB to retrieve relevant information from medical documents and integrated Groq API for response generation.
- Built a React.js frontend and FastAPI backend with MongoDB Atlas for data persistence and efficient user interaction.

**HireMate AI – AI Job Application Assistant** *GitHub | Individual*
*Python, FastAPI, React.js, Machine Learning, NLP, PDF Processing*

- Developed an AI-powered resume and job matching system using Python, FastAPI, React.js, and machine learning to analyze resumes against job descriptions.
- Implemented PDF processing, skill extraction, job-skill matching, missing-skill identification, and personalized recommendations.
- Designed an ATS-focused analysis of resume structure, formatting, skills, keywords, and job-description alignment.

**AI-Based Medical Laboratory System** *GitHub | Individual*
*React.js, Node.js, Express.js, MongoDB, Python, Machine Learning*

- Developed a full-stack AI-powered healthcare platform for disease prediction and laboratory management.
- Implemented machine learning models for multiple disease predictions with secure authentication, REST APIs, and role-based access control.
- Designed appointment booking, medical report management, and personalized health recommendation features.

## Technical Skills

**Languages:** Python, SQL, JavaScript
**Frontend & Backend:** React.js, Node.js, Express.js, FastAPI, Flask, REST APIs, Authentication, RBAC
**Databases:** MongoDB, MongoDB Atlas, ChromaDB (Vector Database), PostgreSQL (Supabase)
**AI/ML:** Machine Learning, Deep Learning, NLP, Generative AI, LLMs, RAG, Semantic Search, Computer Vision
**Libraries & APIs:** TensorFlow, PyTorch, Scikit-learn, NumPy, Pandas, OpenCV, Groq API, Gemini API, Hugging Face
**Document Processing:** PDF Processing, Resume Parsing, Skill Extraction, ATS Analysis
**Tools:** Docker, CI/CD, Git, GitHub, Render, Vercel, Google Colab, VS Code

## Achievements

- **Smart Coder Certification** – Smart Interviews
- **ServiceNow Certified System Administrator (CSA)** – ServiceNow
- **Deep Learning Internship Certification** – CITD, MSME Tool Room, Hyderabad
- **HackerRank Python Certification** – HackerRank